

AI & Machine Learning – Task 4 Report

Classification Models, Evaluation Metrics & Handling Imbalanced Data

Submitted By

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Objective

The objective of this project was to build and evaluate a binary classification model using the Breast Cancer Wisconsin Dataset. The project focuses on understanding classification techniques, evaluating model performance using different metrics, handling class imbalance, and comparing multiple classification algorithms.

Dataset Description

The Breast Cancer Wisconsin Dataset from Scikit-learn was used in this project.

Target Classes

- **0 → Malignant (Cancerous)**
- **1 → Benign (Non-Cancerous)**

The dataset contains medical features extracted from breast cancer cell images and is widely used for binary classification problems.

Data Preprocessing

The following preprocessing steps were performed:

- Dataset Loading
- Train-Test Split (80% Training, 20% Testing)
- Feature Scaling using **StandardScaler**

Feature scaling improves model performance by bringing all features to the same scale.

Logistic Regression Model

A Logistic Regression model was trained as the baseline classifier.

Confusion Matrix

```
[[41  1]
 [ 1 71]]
```

Classification Report

Class	Precision	Recall	F1-Score
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Malignant (0)	0.98	0.98	0.98
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Benign (1)	0.99	0.99	0.99
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Overall Accuracy

98.25%

Observation

The Logistic Regression model achieved excellent classification performance with very high Precision, Recall, and F1-Score for both classes.

Decision Tree Classifier

A Decision Tree Classifier was implemented for comparison.

Classification Report

Class	Precision	Recall	F1-Score
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Malignant (0)	0.85	0.93	0.89
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Benign (1)	0.96	0.90	0.93
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Overall Accuracy

91.23%

Observation

Although the Decision Tree model performed well, its accuracy and F1-score were lower than Logistic Regression.

Model Comparison

Model	Accuracy
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Logistic Regression	98.25%
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Decision Tree Classifier	91.23%
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Best Model

🏆 Logistic Regression

Reason:

- Highest Accuracy (98.25%)
- Better Precision

- Better Recall
 - Better F1-Score
 - More reliable for this dataset
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Confusion Matrix Interpretation

The confusion matrix obtained was:

```
[[41 1]
 [ 1 71]]
```

Interpretation:

- **True Positives (TP):** 71
- **True Negatives (TN):** 41
- **False Positives (FP):** 1
- **False Negatives (FN):** 1

The model made only **2 incorrect predictions** out of **114 test samples**, demonstrating excellent classification performance.

Metric Selection

Accuracy alone is not always sufficient, especially for imbalanced datasets.

Therefore, additional metrics were used:

- Precision
- Recall
- F1-Score
- Confusion Matrix

These metrics provide a more complete evaluation of the classifier's performance.

Handling Imbalanced Data

To address class imbalance, the **class_weight="balanced"** parameter was applied in Logistic Regression.

This technique automatically assigns higher importance to minority class samples, improving fairness and model reliability.

Final Model Decision

Based on all evaluation metrics, **Logistic Regression** was selected as the final model because it achieved:

- Highest Accuracy (**98.25%**)
 - Excellent Precision
 - Excellent Recall
 - Excellent F1-Score
 - Better overall classification performance than the Decision Tree Classifier
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Conclusion

In this project, binary classification techniques were successfully implemented using the Breast Cancer Dataset.

The project included:

- Data preprocessing
- Feature scaling
- Logistic Regression
- Decision Tree Classification
- Confusion Matrix
- Classification Report
- Model Comparison
- Handling Imbalanced Data

The Logistic Regression model outperformed the Decision Tree model and proved to be the most suitable classifier for this dataset.

Tools & Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Scikit-learn
- Jupyter Notebook
- VS Code